

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Information and communications technology

Software Requirement Specification

AIMS

Subject: Software Design and Construction

Group 3

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Hanoi, 12/2024

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1 Introduction

1.1 Objective

This document is the specification of an internet media store, and is tailored for designers, developers and operators to execute this project. Moreover, technical executives can have a grasp of the project for monitoring purposes and users with hope to thoroughly understand the system can also acquire knowledge of the software.

1.2 Scope

Being analyzed in this document is a software called AIMS, stands for An Internet Media Store. As the name suggests, it provides products owners tools to set up and manage their shop online, enables people to buy the merchandise. In executives' point of view, the software is taken advantage as a management tool whose targets are online customers and product manager.

1.3 Glossary

<i>No</i>	<i>Term</i>	<i>Explanation</i>	<i>Example</i>	<i>Note</i>
1	cart	A list of products that the customers want to buy	[CD Lover (1), DVD The on1y one (1)....]	Names are usually followed by quantity

1.4 References

AIMS-problemStatement-v2.0 by Prof. Nguyen Thi Thu Trang

2 Overall Description

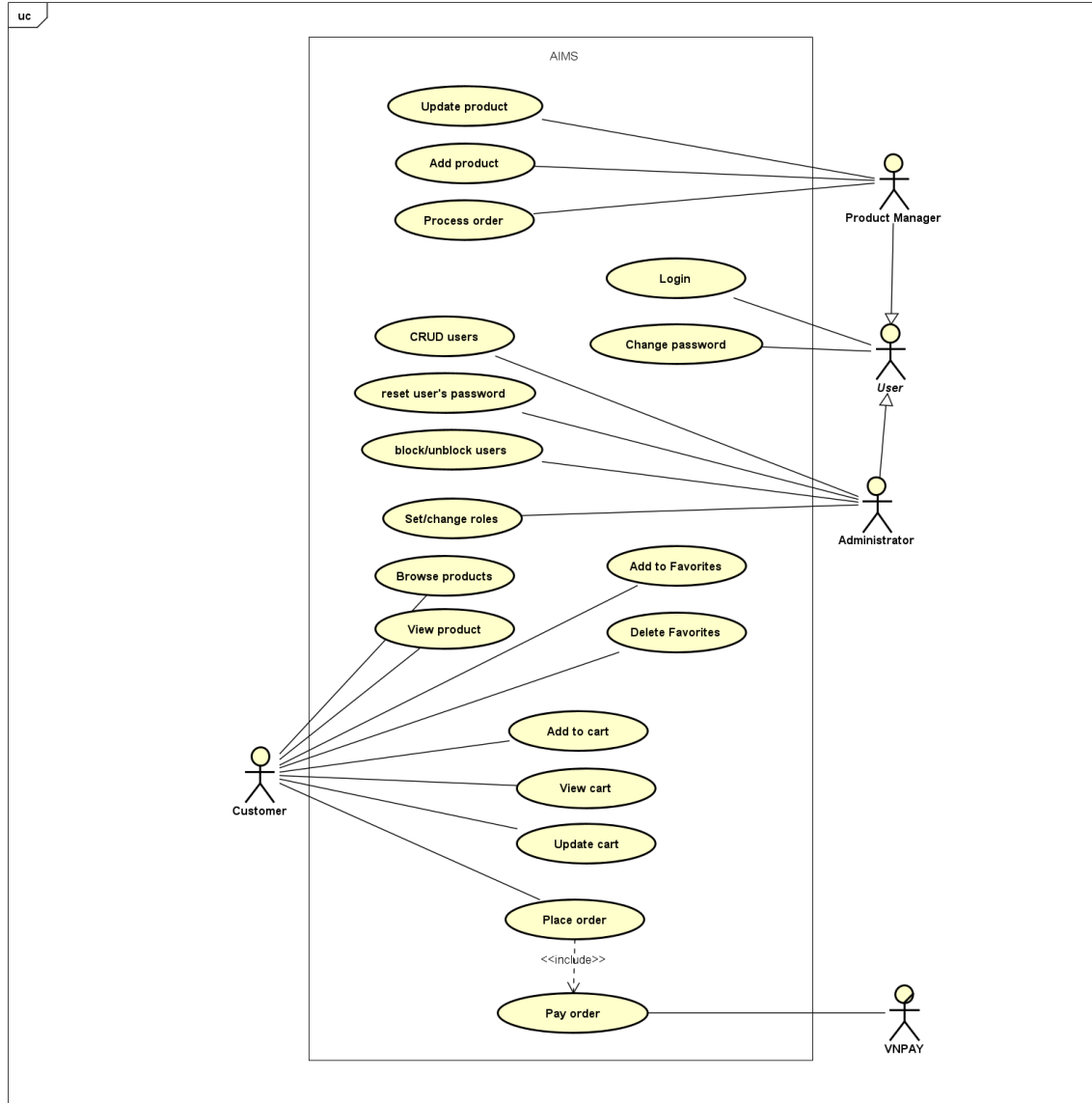
2.1 Survey

AIMS is a desktop e-commerce software that facilitates online shopping, introduces a fast, efficient and secure way of distributing products, specifically, internet media, which includes books, CDs, LP records, DVDs.

Participating in the system, customers, product managers and administrators are the key factors. Customers enjoy online shopping and abide by payment policies, whereas product managers organize their inventories and provide details of the merchandise to the customer. For an order to be accomplished, product managers also need to accept it. Behind the curtains are the administrators who are responsible for managing shops and customers' information, as well as clarifying roles for users of the system. Financially, VNPay joins the process of transacting orders.

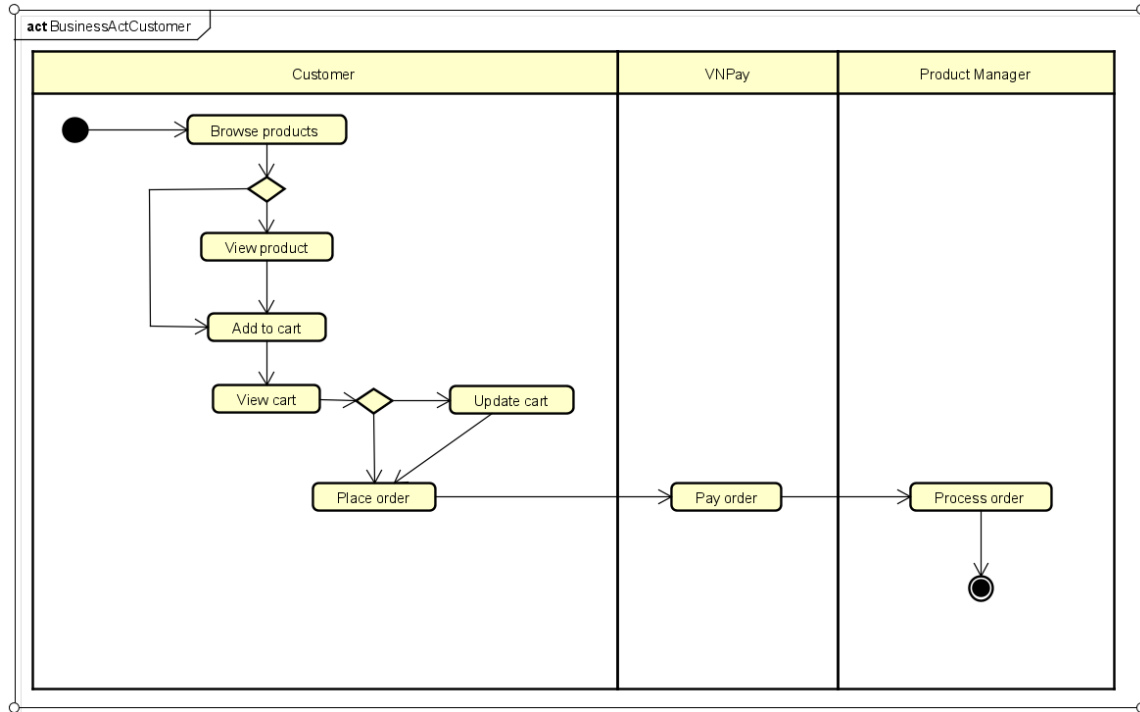
In short, actors involved include Customer, Product Manager, Administrator, VNPay. In order for Administrator and Product Manager to login and be able to change password, an abstract actor called User is added to the system as well.

2.2 Overall requirements



2.3 Business process

In customer's perspective, the sequence of activities can happen in such order described in the following activity diagrams, which shows the process from searching for products to getting order accomplished. As there is no need to login to the system, shopping experience is more simplified; hence, more convenient for users.



As for the administrator, should there be a new sign-up request from either sellers or buyers, they can execute use case Manage users, Manage roles accordingly. At the same time, if the product managers want to update their inventories with item details or quantity, they trigger Update product or Add product use case.

3 Detailed Requirements

3.1 Use case *Place order*

Use Case “Place order”

1. Use case code

UC001

2. Brief Description

This use case describes the interaction between customer and AIMS software when the customer wishes to place an order

3. Actors

3.1 Customer

4. Preconditions

There is at least one item in the cart

5. Basic Flow of Events

1. Customer opens cart, choose the products to place orders
2. AIMS check availability of the products
3. Customer requests to place an order
4. AIMS checks stocks of the products
5. AIMS displays the products with order information
6. Customer chooses delivery method
7. Customer enters delivery information (see Table A)
8. AIMS checks for rush order option
9. AIMS calculates and updates then displays order information and delivery fee (see Table B)
10. Customer asks to pay the order
11. AIMS executes UC “Pay order”
12. AIMS creates and saves a new order, saves the final invoice, send invoice , order transaction information (see table C) through email
13. AIMS empties the cart, displays successful notification

6. Alternative flows

Table N-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
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1.	At Step 3, 5	If the stocks are not sufficient for the order	AIMS notifies the error to customer	Use case ends
2.	At Step 10	If the information is not appropriate	AIMS notifies to the customer, requires customer to refill the information	At step 7
3.	At step 12	If the payment goes wrong	AIMS notifies the error to the customer	At step 9

7. Input data

Table A-Input data of delivery information

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Receiver name		yes		Nguyen Trung Chien
2.	Phone number		yes	10 digits	0123123789
3.	Province	Choose from a list	yes		Ha Noi
4.	Address		yes		1C, Nguyen Van Cu, Long Bien
5.	Shipping instructions		no		Straight to the corner, besides the big fountain
6.	Delivery time	Time and date	no		14:00 20/09/2024

8. Output data

Table B-Output data of order information and cost

No	Data fields	Description	Display format	Example
1.	Title	Title of the product		DVD Lover -TS
2.	Price	Price of the corresponding product	- Comma for thousands separator - Positive integer - Right alignment	313,000
3.	Quantity	Quantity of the corresponding product	- Positive integer - Right alignment	1

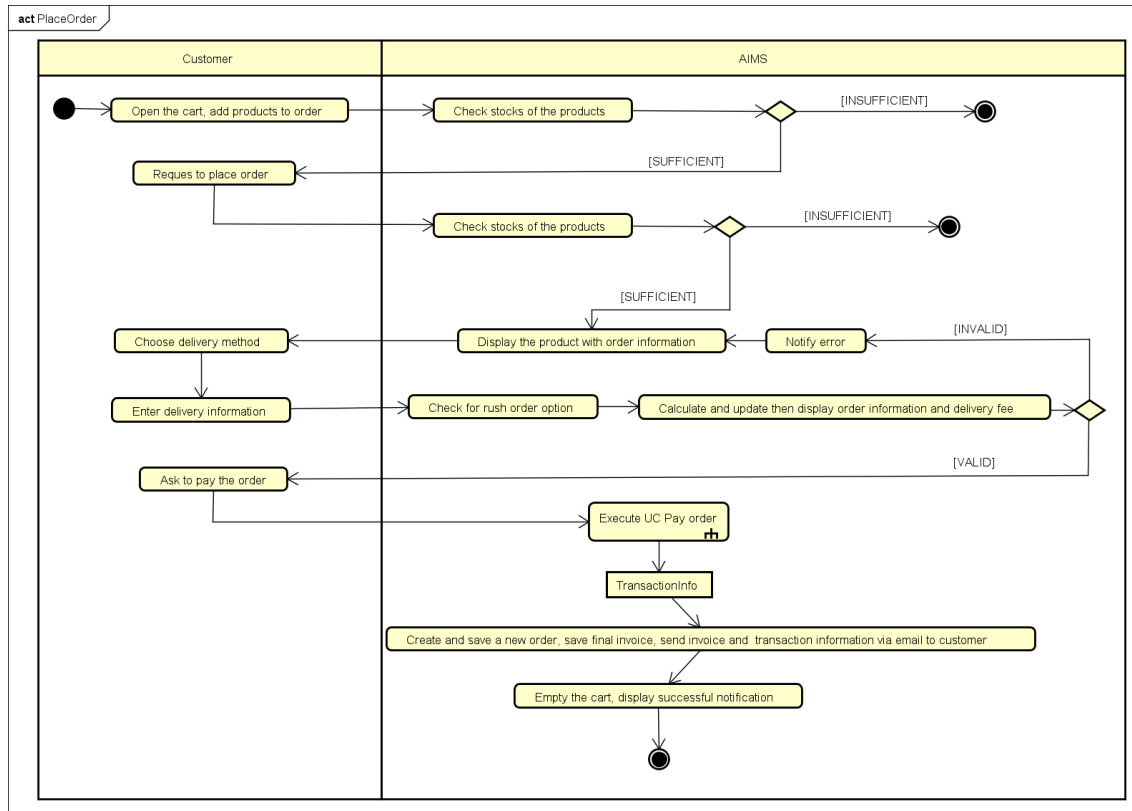
4.	Amount	Total money of the corresponding product	- Comma for thousands separator - Positive integer - Right alignment	313,000
5.	Subtotal	Sum cost of the cart		512,000
6.	Ship	Delivery fee		15,000
7.	Total	Total cost including shipping fee		527,000

Table C-Output data of order transaction information

No	Data fields	Description	Display format	Example
1.	Customer name			Nguyen Thinh Vong
2.	Phone number			0123123789
3.	Province			Ha Noi
4.	Address			1C, Nguyen Van Cu, Long Bien
5.	Total	Total cost of the order	- Vietnamese currency - Comma for thousands separator - Positive integers - Right alignment	527,000 đ
6.	Transaction ID			AIMSXP00123123
7.	Transaction content			
8.	Transaction date		dd/mm/yyyy	04/12/2024

9. Postconditions

A new order is created, its information is sent via email to the customer, AIMS notifies successful transaction. If the payment is not successful, nothing happens



3.2 Use case Pay order

Use Case “Pay order”

1. Use case code

UC002

2. Brief Description

This use case describes the interaction between AIMS software and VNPay when the customer initiates paying order

3. Actors

3.1 VNPay

3.2 Customer

4. Preconditions

Customer requested order payment

5. Basic Flow of Events

1. AIMS software redirects payment process to VNPay
2. VNPay display total cost that needs paying

3. Customer asks to pay
4. VNPay executes payment
5. VNPay sends payment result to AIMS
6. VNPay sends transaction information to AIMS
7. VNPay notifies payment result

6. Alternative flows

Table N-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 5	If the customer cancels payment or error exists	▪ AIMS notifies the situation to customer, refunds through VNPay	Use case ends

7. Input data

8. Output data

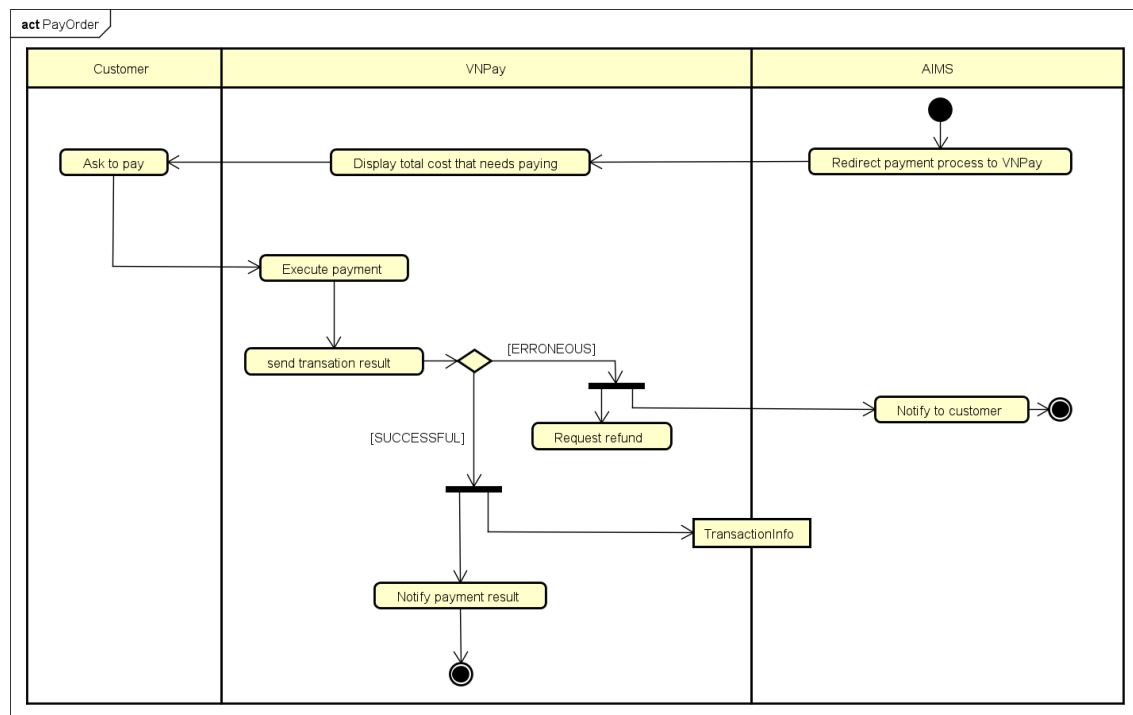
Table A-Output data of invoice information

No	Data fields	Description	Display format	Example
1.	Title	Title of the product		DVD Lover -TS
2.	Price	Price of the corresponding product	▪ Comma for thousands separator ▪ Positive integer ▪ Right alignment	313,000
3.	Quantity	Quantity of the corresponding product	▪ Positive integer ▪ Right alignment	1
4.	Amount	Total money of the corresponding product	▪ Comma for thousands separator ▪ Positive integer ▪ Right alignment	313,000
5.	Subtotal before VAT	Sum cost of the cart before VAT		502,000
6.	Subtotal	Sum cost of the cart with VAT		512,000
7.	Ship	Delivery fee		15,000
8.	Total	Total cost including shipping fee		527,000
9.	Currency			đ
10.	Name			Nguyen Trung Chien

11.	Phone number			0123123789
12.	Province			Ha Noi
13.	Address			1C, Nguyen Van Cu, Long Bien
14.	Shipping instructions			Straight to the corner, besides the big fountain

9. Postconditions

An invoice of the order is sent to AIMS, VNPay notifies payment results. If the customer cancels the order, VNPay notifies refund information.



4 Supplementary specification

4.1 Functionality

Customers can add products to the cart whenever the product is shown to them, regardless of which screen is being displayed. Whenever landing show-products status, the software will display the first 20 items, which are either random or related to customer's wish.

4.2 Usability

The system does not require login to be able to purchase a product, each transaction will be considered as one working session. Customers can sort the list of products by price, and view each product's details. Users of other online shopping platforms should not be estranged from the process

4.3 Reliability

The software assures the price of a product stays between 30% and 150% of its value. Product managers are able to add product information and update stocks up to 10 products at once and only up to 30 products per day. Customers can only delete 1 product from their cart at a time. Product managers have to verify order from customers. Payment will be processed with VNPay, any problem occurring during the transaction would cancel the payment and trigger refunding to customers if necessary.

Meantime between failure = 300 hours.

4.4 Performance

The application operates 24/7, can serve up to 1000 customers simultaneously without significant reduction in performance. Maximum response time of the software is 2 seconds under normal conditions or 5 seconds during peak hours.

4.5 Supportability

If any problem or failure is to happen, the software can resume to normal operation within a maximum of 1 hour afterwards. Administrators and Product Managers have to login to access their management tools, as well as change their passwords. If a user is to reset their password, he/she can request administrators to get the task done. Any changes to customers' data should be notified directly to the screen and suggested actions if available.

4.6 Other requirements

Connection to VNPay Sandbox should be implemented for payment development.